
CSE 4303 Final Project – An Exploration of Data Remanence

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Video: [FinalVideo.mp4](#)

1. Project Overview

1.1. Background

Data Remanence is the idea that deleted information remains on storage devices in spite of simple deletion, which just removes an entry from the filesystem. This concept is useful in fields such as digital forensics, data recovery, and media sanitization. This project aims to explore data remanence, its relevance in various storage devices, and the policies that within electronic recycling that consider data remanence. The aim is to address the confidentiality pillar of computer security, as the ability to uncover information previously thought to be removed could potentially lead to sensitive information being found. For example, in hospitals, when they get new devices, the old ones need to be properly erased before being trashed or repurposed, as to not potentially leak patient information. Understanding the current tools for uncovering deleted information, and the policies for the recycling of electronics can properly prevent such leaks from happening.

1.2. Project Objectives:

1. Garner an understanding of how different storage devices handle deletion in common storage devices (flash drives, hard drives) and less common ones (MP3 players, Nintendo DSi, Old Phones, Cameras).
2. Explore how to recover deleted file information on various devices.
3. Learn current electronic recycling policy, and how it aligns with remanence of data on devices.

1.3. Plan:

- Week 1-2: Topic selection + background
- Week 2-3: Tool experiment
- Week 5: Project write up and Presentation recording

2. Existing Tools and Literature

2.1. Data Remanence

Data remanence is a well-researched topic. Ever since computers first had the ability to store memory, there has been research regarding how to properly clear that memory. There exist many standards regarding how sanitize storage devices such as the IEEE Std 2883-2022 and NIST SP 800-88r2. These standards are useful in terms of preventing data remanence in semiconductor devices such as flash drives and magnetic storage devices such as hard drives. More modern literature studies remanence within RAM and cloud storage. RAM related data remanence literature has to do with the ever-increasing use of embedded systems allowing for cold boot attacks on DRAM, and potentially volt boot attacks on on-chip SRAM [1]. Regarding the cloud storage research, cloud storage does not localize files to one location as on computer storage might, rather dispersing one's files among several servers. As such, it is harder to ensure that something deleted is gone from every server the file could be actually stored on. Due to this there is current research regarding developing policy for preventing data remanence in cloud storage [2][3].

Additionally, there are preexisting tools for both uncovering and erasing information from devices. In this project, the tools FTK Imager, The Sleuth Kit, Autopsy, and iLeapp were used to recover information from storage devices. FTK Imager is an application that takes forensic images, which is essentially a copy of all data on the device [4]. Using an image ensures that none of the original data is changed. Then the tool Autopsy was used to parse the images. Autopsy is an open-source digital forensics application that is essentially a GUI for The Sleuth Kit. It scanned the image for all information within the filesystem or deleted and has the ability to sort out from the image deleted files, emails, browser history, installed applications, and flag any files deemed “suspicious” based on inputted parameters and file metadata [5][6]. iLeapp is an open-source tool for parsing IOS filesystems, something FTK Imager cannot do in its free version [7].

2.2. Electronic Recycling

Regarding electronic recycling, there is not much standardization. In the United States, electronic recycling policy is a state regulated issue, with only 25 states having laws regarding e-waste [8]. These laws vary, with some regarding the individual consumer, and others having to do with schools, businesses, and government entities in the state. In the state of Missouri, legislation was passed in 2008 titled the “Manufacturer Responsibility and Consumer Convenience Equipment Collection and Recovery Act”, which requires the manufacturer and retailers of electronics to be responsible for the collection and recycling of the electronics they distribute [9]. This act, as well as the ones in the other 24 states, does not cover data on the devices, just the recycling of the devices themselves. Secondhand distributors such as eBay, Facebook Marketplace, and Goodwill have no rules regarding how one removes information from electronic devices they are trying to resale. In terms of flash drives, hard drives, and SD Cards, many may assume that deleting the files is enough. For computers and phones, many just do a factory reset before selling their device, which is mostly effective, but there have been previous cases where factory resets still had data remanence issues. A study from 2017 displays a factory reset flaw in Android devices and shows how a substantial amount of information could still be uncovered post reset [10]. While it has been a significant amount of time since that paper’s publication, it is important to consider that older devices being redistributed running older OS’s might still contain these flaws. Overall, data remanence is not explicitly considered in much of electronic recycling and redistribution.

3. Experiment Write up

3.1. Recovering Information

The first step in this exploration of data remanence was to see how to recover information, both visible and deleted, from various devices. This process began by collecting various storage devices. Fortunately, all used were either already owned or gifted, so no money was spent on used devices. The devices that were used are seen in Table 1.

After collecting the devices, images of each device were taken using the free version of FTK Imager. For the iPhone, a regular image could not be taken in a similar method using FTK Imager, so an iTunes backup of the phone was made, and iLeapp was used to analyze the backup. For the other images, the tool Autopsy was used to analyze each storage device. Table 1 displays the initial files seen on the devices, the files uncovered from said devices, and the information gathered from those files.

Device	Visible Files on File Explorer	Total Uncovered Files	Information Revealed
PNY 32G Flash Drive	9	97	PowerPoint Presentation from a Lego Mindstorm Competition, A resume document

WD Elements Hard Drive	0	19373	GoPro recordings of Bike rides, Biology class videos and photos, Album cover art
iPhone 5s*	8.15Gbs	-	Contact lists, Apps on the phone, Internet search history, call history, and all preference settings.
SD card of Samsung Galaxy S2	20	11	Photos and videos of pets
2012 MacBook SSD	8.15Gb		This device was reformatted to run Debian
SD Card of Cannon EOS Rebel T3i	2.2Gb	99	Filenames of deleted photos
iPod Nano 2 nd Gen	10	45	Deleted Mp3 files
iPod Nano 3 rd Gen	24	9	Deleted Mp3 files
EVO 32G Flash Drive	49	96	Videos from events
SD Card of Nintendo DSI	178	47	Deleted Mp3 files

Table 1: A table displaying the devices, and their visible and deleted files. It should be noted, the iPhone 5s's files were uncovered using iLeapp rather than Autopsy, so it does not explicitly highlight uncovered deleted information.

Much of the information on these devices could be potentially sensitive. For devices such as the two iPods, there isn't much concern about sensitive information sans obtaining someone music listening habits, but in the EVO 32G Flash Drive's video footage from events, the SD Card of Cannon EOS Rebel T3i's deleted images, and the WD Elements Hard Drive video footage of bike rides, are all information that could be potentially sensitive. The iPhone 5s contains information that has the strongest potential to be sensitive, as through iLeapp, contact lists, call and search history, and setting preferences were all able to be parsed.

3.2. Exploring Policy

Electronic recycling, as a general topic, is regulated state-by-state, with only half of the states in the U.S. having laws regulating the recycling of electronics. These policies, along with the policies of 2nd-hand sellers, do not concern themselves with data remanence on such devices. Laws regarding electronic recycling are concerned with the preventing generation of e-waste. In the state of Missouri, it is up to the distributors and manufacturers to handle recycling, and it is also illegal for group such as schools and government entities to put their electronics up for recycling. This is as far as the concern for data remanence goes.

As more electronics get released and, as a result, more electronics get tossed out or sold, it is important to consider the information on the current everyday devices of individuals, businesses, schools, and more that current policy doesn't. While Missouri and other states leave it up to the manufacturers and distributors to offer recycling plans, that doesn't mean the individual has to utilize them. A current listing on Facebook Marketplace, is selling computer parts with the description stating that the seller has "bins of older medical equipment computer parts," [11]. There is no guarantee that this seller has properly wiped these memory storage components they are selling properly, meaning there could be sensitive information on them.

Ultimately, this begs the question of who is responsible for the remanent data on storage devices. Companies that offer e-recycling services often also offer data removal services, but these must be explicitly requested. If one donates a bag of old flash drives or SD cards from a school or business, and they don't remove any information from them, just go through the proper means of disposal, there is no

guarantee those flash drives are free of sensitive information. If one personally sells an old device, or donates it to a secondhand shop, they must personally handle the removal of sensitive information. Since it is up to the individual or business to handle the proper sanitization of their data, it does make one wonder if the average person is knowledgeable enough on their remanent data to properly remove any sensitive information. Current studies on public perceptions regarding e-waste don't consider the existence of sensitive information on the devices, but rather a more general disposal of electronics [12][13][14]. There is a clear gap in research regarding how the public perceives their data on old devices, and policy for e-waste regarding remnant data.

4. Conclusion

This project had the general goal of exploring data remanence in storage devices from different sources. A total of about 19.5K deleted files from 10 separate devices was uncovered, all with varying levels of sensitivity, from deleted music files to call histories. It shows overall how different storage devices can hold information thought to be deleted, and how those files can be easily uncovered with relatively simple, open-source tools.

The secondary part of this project investigated the current e-recycling policies, and how they, or rather how they do not, consider data remanence. It was found that much electronic policy is concerned with proper disposal of electronic waste, and not the information that could be held on those devices. The onus is on the individual for dealing with the data on the devices they are recycling.

In terms of next steps, it would be interesting to explore public perceptions towards data remanence in recycled /resold devices. If sellers of electronic on websites such as Facebook Marketplace or eBay were surveyed, how would they consider data remanence on the things they are selling. There are already standards for the sanitization of storage devices, but do these sellers know of them and follow them? It would also be interesting to explore remanence in cloud storage, as that mitigates much of the on-device sensitive information, but when people delete information from their cloud storage, how is one sure that it is deleted? There has been research on this topic, but it is all from a few years ago, and I think looking at the many different cloud storage options, and how proposed deletion policies hold currently could be interesting.

5. References

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